

Marking Scheme

- 1** A DNA library is a collection of DNA fragments that have been cloned into vectors so that researchers can identify and isolate the DNA fragments that interest them for further study. There are basically two kinds of libraries: genomic DNA and cDNA libraries.

(a) Outline how a cDNA library is produced.[2]

Reverse transcriptase uses the mRNA isolated from a specific tissue as a template to reverse transcribe to form a single-stranded cDNA copy;

RNaseH is added to partially digest the mRNA strand and DNA polymerase is used to initiate synthesis of a complementary cDNA strand;

Terminal transferase adds linkers to the resultant double-stranded cDNA to allow the cDNAs are inserted into plasmids.

Recombinant plasmids are then introduced into competent bacterial host cells to be stored to form the library;

(b) Explain why the cDNA library is preferred over the genomic library by the researchers. [3]

The cDNA library is smaller OR more compact as it only contains only the coding sections of DNA OR exons;

Introns removed OR no junk DNA present (Reject: non-coding genes absent) thus it is easier to find desired gene

Genes present in clones are those that are being expressed OR reflects proteins actually represented in the cells;

AVP;

The presence and absence of the restriction site generates a restriction fragment length polymorphism (RFLP). RFLP can be a useful biological tool, for example as a marker in chromosomal mapping.

(c) State **one** other ways in which RFLP can be used as a biological tool.[1]

Disease detection OR DNA fingerprinting

A: paternity testing / family tracing as these are subcategories for DNA fingerprinting

R: forensics (vague)

(d) Outline how the relative positions between the particular gene and the two RFLP markers can be determined.[3]

The order and the relative distances between the RFLP markers and the disease gene on a linkage map are based on recombination frequencies obtained.

The recombination frequency is a measure of genetic distance between two loci on a DNA molecule / OWTTE

The closeness of the RFLP marker A, B and the disease gene can be measured by:

- The frequency at which two RFLP markers (A and B) are inherited together OR
- The frequency at which the disease gene and RFLP marker A are inherited together OR
- The frequency at which the disease gene and RFLP marker B are inherited together

RFLP analysis is often applied to genomic DNA after digesting with restriction enzymes and separating the DNA by gel electrophoresis. Gel electrophoresis of this DNA produces a smear of DNA, rather than distinct bands.

For a smear of digested DNA, additional steps involving DNA hybridisation are needed to analyse the RFLP pattern for a particular gene.

- (e) Outline the process of DNA hybridisation that allows the RFLP pattern for a particular gene to be visualised.[5]

The double helix structure of the DNA is drawn from the gel onto a nitrocellulose membrane by a blotting process.

The double stranded DNA on the nitrocellulose membrane is then split by the addition of an alkali, e.g. NaOH.

Radioactively-labelled / fluorescence-labelled probes are added.

Thus the probe will bind to the complementary sequence of single-stranded DNA on the nitrocellulose membrane.

The membrane is then placed next to a photographic film and the genetic fingerprint can be seen when the film is developed. [for radioactive probe] / The membrane is then placed under a UV lamp and the genetic fingerprint can be seen. [for fluorescence probe]

[Total: 14]

- 2 Telomere length regulates the life span of a cell, in which the numbers of cell cycles are limited before the cell undergoes cell senescence. Thus, telomeres limit the capacity of a cell to divide. Fig. 2.1 shows the relationship between telomere length and the number of cell cycles for the different types of cells in the body.

Telomere length

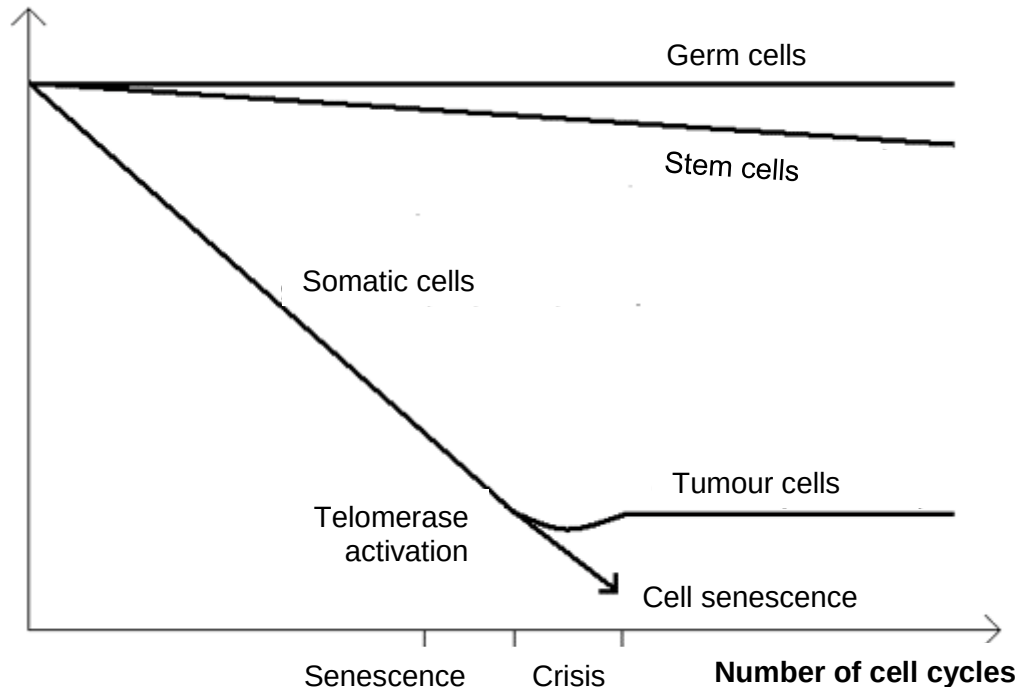


Fig. 2.1

With reference to Fig. 2.1,

- (a) state one similarity in the relationship between telomere length and the number of cell cycles in stem cells and tumour cells. [1]

Both stem cells and tumour cells can divide by mitosis **many times as telomere length is longer than hayflick limit / critical length.** / OWTTE

A: higher number of cell cycles, telomere length remains constant for both types of cells.

- (b) explain how the relationship between telomere length and the number of cell cycles support one of the properties of stem cells. [2]

This is because the telomere length in stem cells **decreases slowly / erodes at a relatively slower rate / is maintained at a relatively constant length** thus allowing them to replicate many times to produce more daughter stem cells.

Hence, stem cells are able to **self-renew / self-regenerate / divide continuously by mitosis** for a longer period.

Cytokines bind Type I/II cytokine receptors that have a common receptor subunit called the common gamma-C chain (γ_c). Mutations of γ_c gene underlie X-linked severe combined immunodeficiency (X-SCID) and account for roughly half of all known cases of SCID. Fig 2.2 shows that the deficiency of γ_c protein blocks signalling by IL-7 and IL-15 which is required for haematopoietic stem cells to develop into natural killer (NK) cells and T cells. NK cells and T cells are lymphocytes that play critical roles in the immune system.

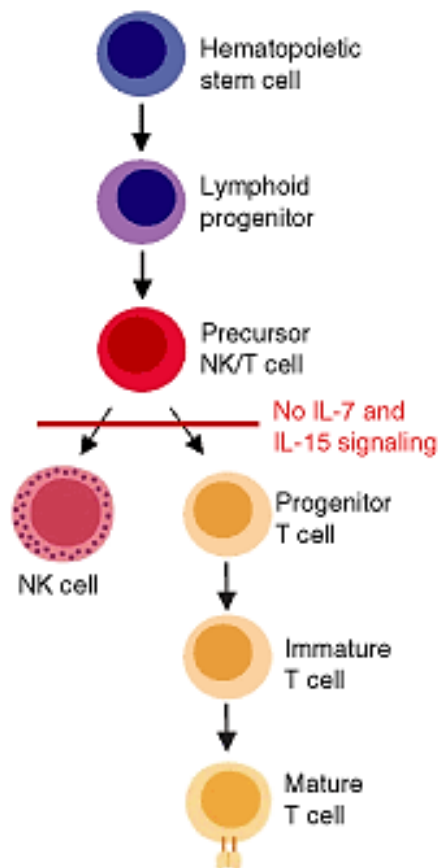


Fig. 2.2

- (c) With reference to the information given and Fig. 2.2, explain why individuals with X-linked SCID have high mortality. [2]

Individuals with X-linked SCID are **unable to express gamma-C chain gene** to produce function IL-7 and IL-15 thus **signalling by IL-7 and IL-15 is blocked**.

This **prevents the differentiation of precursor NK/T cell** into functional / mature NK and/or T-cells, thus individuals have weaker immune system which makes them **susceptible to many infections that can cause death**.

Gene therapy has been used to treat patients with X-linked SCID. X-linked SCID is a recessive disease. Gene therapy uses a vector to deliver a normal copy of a γ c gene to the cells where it is needed.

- (d) Describe **one** non-viral gene delivery system that could have been used to insert normal copy of a γ c gene into stem cells. [4]
1. Use an appropriate restriction enzyme to excise the **normal allele** that codes for gamma-c and to linearise an appropriate bacteria plasmid. Use DNA **ligase** to form the phosphodiester bonds between adjacent nucleotide sequences to form the recombinant plasmid;
 2. Insert the recombinant plasmid into a **liposome** that is coated with specific receptors, such that the liposome will identify and adhere to the stem cells;
 3. Allow the plasmid-containing liposome to get in **contact** in-vitro with the stem cells to be treated;
 4. Upon membrane fusion between the phospholipid bilayers of the liposome and stem cell surface membranes (**A: receptor-mediated endocytosis**), the recombinant plasmid will be released into the nucleus for the stem cell genome to take up the normal gamma-c allele via homologous recombination;

A: gene gun (if student is able to elaborate)

1. Use an appropriate restriction enzyme to excise the **normal allele** that codes for gamma-c and to linearise an appropriate bacteria plasmid. Use DNA **ligase** to form the phosphodiester bonds between adjacent nucleotide sequences to form the recombinant plasmid.
2. Gold or tungsten spherical particles are coated with plasmid DNA.
3. Metal particles are accelerated to high speed by pressurised gas in the gene gun into stem cells.
4. The recombinant plasmid will be in the nucleus for the stem cell genome to take up the normal gamma-c allele via homologous recombination;

- (e) Explain why it is sufficient to insert a single normal copy of a γ c gene into stem cells for gene therapy.[2]

The normal gamma-C allele is dominant;

Hence one copy of the dominant allele is sufficient to produce functional protein products;

[Total: 11]

- 3** The *Aedes aegypti* mosquito is the main vector that transmits the viruses that cause dengue. Brazil is one of the first few countries to attempt to reduce the numbers of *A. aegypti* by using genetically modified (GM) male mosquitoes. This project was initiated a year after an epidemic of dengue fever that caused more than 1.5 million cases in Brazil.

The male *A. aegypti* mosquitoes are genetically engineered to contain a *tTA* gene that produces a protein called *tTA*. When they are released into the environment and mated with the female mosquitoes, the *tTA* gene is passed to the offspring, causing death. This will bring down the population numbers.

A heritable, fluorescent marker (GFP) gene was also introduced into the engineered mosquitoes to distinguish them from native pest insects and to help scientists with the management of pest control programmes.

- (a)** Explain why, in many examples of gene technology, fluorescent markers are used in preference to antibiotic resistance genes.

- easier to, identify/screen for transformed cells as only U.V light is required.
 - more economical/time saving/labour saving with an reasonable explanation like no need to replica plate
 - resistance gene(s) can be passed to other bacteria;
 - idea of antibiotics no longer effective
- OR
- requiring development of new, antibiotics/treatments; [1]

Fig. 3.1 shows the transgene that has been added to the GM mosquito.

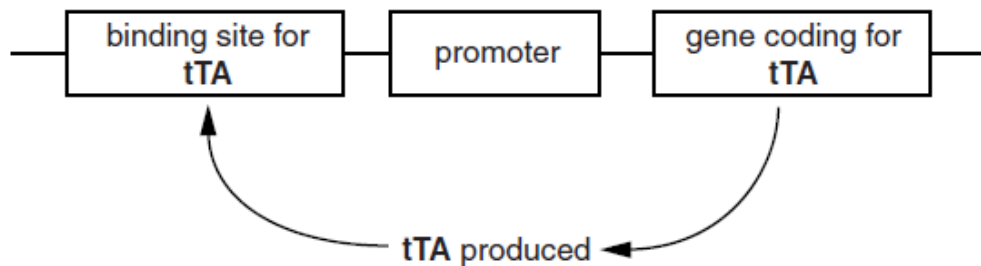


Fig. 3.1

- (b)** With reference to Fig. 3.1, explain why a promoter needs to be transferred along with the *tTA* gene.

RNA polymerase and general transcription factors bind at the promoter to initiate transcription / switch on gene expression;

If the promoter is not transferred along with the desired gene, the gene has to be inserted near an existing promoter, this may disrupt expression of existing gene;

in eukaryotes precise position of promoter is important for gene expression; [2]

In addition to GM mosquitoes, there are experiments using mosquitoes infected with a bacterium, *Wolbachia*, that seems to prevent them from spreading diseases.

The National Environmental Agency (NEA) of Singapore is conducting studies on the feasibility of using bacteria, *Wolbachia*. The bacterium is found naturally and used in the laboratory to infect male *A. aegypti* mosquitoes.

Wolbachia-infected male *A. aegypti* mosquitoes are released to mate with the females and their eggs do not hatch. This helps to suppress the Aedes mosquito population in Singapore.

- (c) Describe reasons why some members of the public will prefer to use mosquitoes infected *Wolbachia*, instead of using genetically modified mosquitoes to decrease the population of Aedes mosquito.

Not genetically modified, thus the use of infected mosquitoes will not have the concern of foreign genes inserted into another species.

Tampering with the genome of another species is going against nature (accept alternative words)

The bacteria is found naturally, there is no worry of the introducing novel genes into the environment.

With the equipment and specialised skills required for transgenic technology, the cost of genetically modifying the mosquito may be more expensive than infecting the mosquitos with a bacteria that is found naturally. [2]

- (d) The two methods of controlling mosquito population have some disadvantages. Suggest one disadvantage.

Both methods will involve the continuous/repeated release of modified males because the genetically modified or the infected males will not leave any viable offspring to reproduce; costly/dependent on the company/ lock in with company for long periods.

Both methods will involve the release of large numbers of modified mosquitoes as the chances of a female mating with a modified male must be higher for this method to work; costly.

Genetic engineering can also be used to make transgenic plants that increase the yield of cash crops. Fig. 3.2 shows the procedure of transforming *Brassica* using this method.

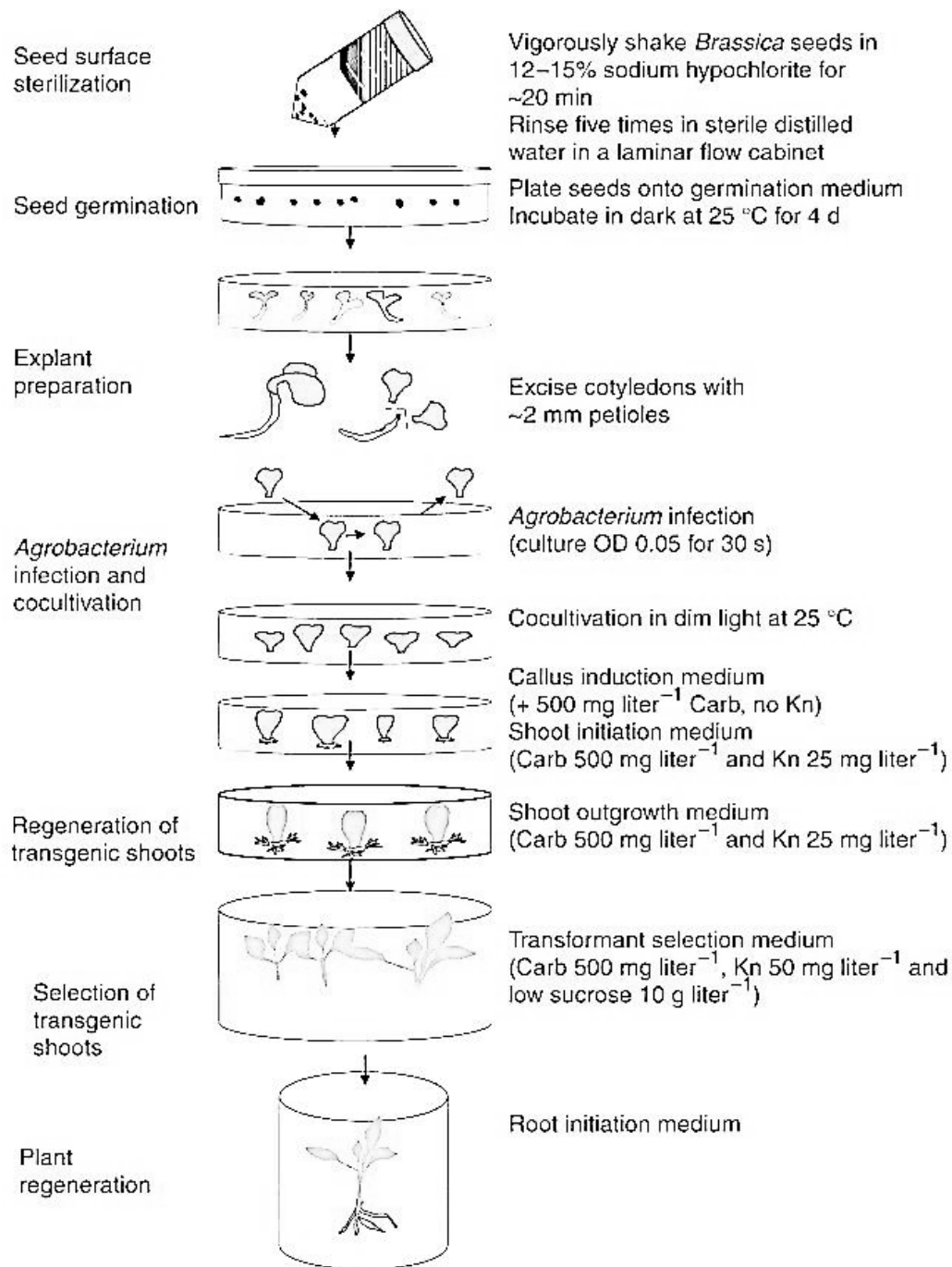


Fig. 3.2

- (e) With reference to Fig. 3.2, explain why carbenicillin and kanamycin were added in the various media.
- Carbenicillin and kanamycin are antibiotics, which were added to select for the growth of transformed Brassica;
 - OR
 - Carbenicillin and kanamycin are antibiotics; carbenicillin is added to inhibit the growth of *A. tumefaciens* attached to the explants and kanamycin added subsequently to select for the growth of transformed Brassica;
 - The genetic markers on the plasmid are therefore genes encoding for resistance to carbenicillin and kanamycin;
 - Only successfully transformed cells/ plantlets will survive in culture; [3]
- (f) Explain why plant tissue culture technique is used to regenerate the transformed plants.
- They have the advantage of being free from fungal or bacterial infections because the PTC is done in a sterile/pathogen free environment;
 - Quick production of mature plants;
 - Regeneration of whole plants from genetically modified explants;
 - Many Genetically modified plants can be grown from one a few plant cells
Amount of space required for field trials to determine which plants have desirable traits is reduced; [2]
- (g) State the difference in composition between the root initiation medium and the shoot outgrowth medium.
- Root initiation medium: high auxin:cytokinin whereas;
Shoot outgrowth medium: high cytokinin:auxin [1]
- (h) Explain the significance of genetic engineering in improving the quality and yield of crop plants and animals using **named examples**.

Must include one example of plant and animal each to get the full mark, if both examples are of plants/animals, a max of 1 mk is given

Golden Rice can help to **alleviate vitamin A deficiency** due to the insertion of a gene that encodes for **beta carotene (precursor to Vitamin A)**. rice becomes **more nutritious (increase in quality)** → decreased malnutrition in underdeveloped countries.

Bt corn/ cotton contains **a gene from a naturally occurring soil-borne bacterium, *Bacillus thuringiensis*** that can kill **insect pests that feed on the plants**. This reduced reliance on chemical insecticides → in turn reducing cost, labour to employ staff to spray chemical insecticides, environmental damage (Bt toxin breaks down rapidly in the environment)/ Higher yield → due to reduced crop losses

Transgenic salmon carries a **gene that codes for the production of Chinook salmon growth hormone, thus increasing their growth greatly. Antifreeze proteins (AFPs)** function to lower the freezing temperatures of fish blood and extracellular fluids to protect fish from freezing in frigid marine waters. •

Faster growth rate means salmon farmers are able to market their fish earlier, and this means increased overall production and profits due to a faster production cycle.

Other revelant NAMED examples. [2]

[Total: 15]

Suggested answers**Background theory: max 3 marks**

- In excess substrate, hydrogen peroxide, as catalase enzyme concentration increases, **more active sites are available**.-->More enzyme-substrate complexes are formed. Hence higher rate of reaction. Result in higher production of oxygen from decomposition of hydrogen peroxide.
- At high concentration of catalase , **not enough substrate molecules to occupy all the active sites** of the enzyme molecules.
- Rate of reaction limited by enzyme concentration/ A point is reach when further increase in enzyme concentration **does not increase** the rate of reaction.--> **plateau** in rate of reaction.
- Increase in oxygen produced measured by change in pressure and can be read from a pressure sensor

Variables

Dependable variable : The change in pressure in kPa over 1 min to obtain initial rate in kPa min⁻¹ / 30 s to obtain kPa s⁻¹

Independent variable : enzyme concentration in % /mol dm⁻³ (5 ranges)

Fixed variable : Volume and concentration of hydrogen peroxide. Temperature , pH , mass of potato AVP (state at least 3)

Methodology:

- **Preparation of catalase concentrations**

1. Potato peeled, cut , grind with buffer, filter with cheese cloth to obtain potato liquid containing catalase
2. Prepare 5 different concentrations of catalases solutions according to the dilution table shown below. The volume of each concentration will be the same eg 10 cm³.

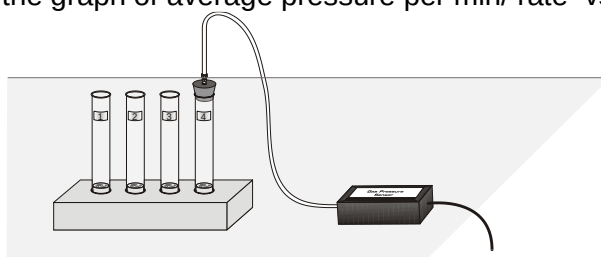
Concentration of catalase (%)	Volume of stock catalase solution (cm ³)	Volume of distilled water (cm ³)
10.0	10.0	0.0
8.0	8.0	2.0
6.0	6.0	4.0
4.0	4.0	6.0
2.0	2.0	8.0
0.0	0.0	10.0

[2]**Procedure**

1. Pilot test*Conduct a pilot experiment to determine suitability of apparatus, suitability of range of independent variable (e.g. enzyme range), optimum conditions, amount of materials used (number of pieces of potatoes, volume of distilled water)
2. Place 5 test tubes in a rack and label 1, 2 etc
3. Fill each test tube with 5 cm³ of 3.0% H₂O₂
4. Place all tubes in water bath of 38-42°C water bath for 2 min
5. Acclimatize the catalase solutions separately, in a 38-42°C water bath for 2 min.
6. After 2 min, using a clean syringe, add 0.5 cm³ of enzyme suspension (10%) to test tube 1.
7. Mix well by stirring and connect the Gas pressure Sensor to the computer interface. Start time. Click to begin data collection.

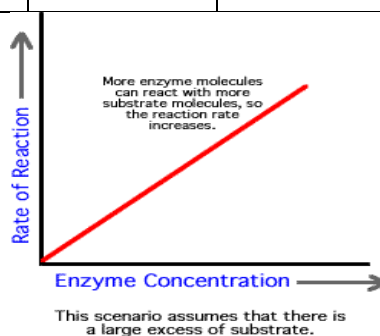
Allow the reaction to take place for 1 minute/ 30 s

8. Immediately after 1 minute, read the pressure of the oxygen gas formed and obtain change of pressure.
9. Collect data for test tubes 2 and 3 till 5 for the different catalase concentrations.
10. **Repeat Steps 2–10** 3 times to obtain replicates in order to obtain **average readings** and ensure no anomalies.
11. Repeat **preparation of catalase suspension and dilution** and **repeat steps 1 to 11** at least 2 times to ensure reproducibility
12. **Control:** Replace catalase with distilled water and add hydrogen peroxide.
Purpose: To show that any production of oxygen still due to the presence of catalase which cause decomposition of hydrogen peroxide, and hence the results obtained due to presence of catalase [1]
13. Obtain the graph of average pressure per min/ rate vs enzyme concentration



must annotate correctly

Concentration of catalase / %	Rate / kPa min ⁻¹			
	Replicate 1	Replicate 2	Replicate 3	Average
10.0				
8.0				
6.0				
4.0				
2.0				
0.0				



Risk assessment [1] at least 2 correct

Risk	Precaution
Glassware	Breakage → clean up carefully to avoid cuts
Use of electric blender	Ensure hands dry when switching on the power to prevent electrocution
Hydrogen peroxide	Corrosive – wear goggles and gloves

5 Free response question

Write your answers to this question on the separate answer paper provided.

Your answers:

- should be illustrated by large, clearly labelled diagrams, where appropriate,
 - must be in continuous prose, where appropriate,
 - must be set in sections **(a)**, **(b)** etc., as indicated in the question.
- (a) Explain the problems associated with the expression of eukaryotic genes in prokaryotes and how these problems are overcome. [6]
1. Prokaryotes (bacteria) lack: ability to perform post-transcriptional modifications → they lack the RNA splicing machinery, thus they are unable to remove the introns of an eukaryotic gene.
 2. Also, with introns, the eukaryotic gene can be too long to be incorporated into the plasmid.
 3. mRNA with introns removed are used– using reverse transcriptase → form cDNA complementary to RNA in 5' → 3' direction.
 4. Then mRNA is removed and another complementary DNA is synthesised by DNA polymerase using the single stranded DNA.--> blunt ends DNA being the eukaryotic gene. Then form recombinant DNA to transform into bacteria to clone the gene when bacteria undergo binary fission.
 5. Prokaryotes (bacteria) lack: ability to perform many post-translational modifications → they cannot produce a functional protein and any example add disulphide bridges to join polypeptide A and B in insulin → unable to fold into 3-D configuration that is functional;
 6. form artificial gene for each polypeptide and form recombinant DNA separately. Transform into bacteria and form fusion proteins separately. Extract, purify and to allow formation of bonds in-vitro
 7. Prokaryotes (bacteria) lack ability to recognise eukaryote's promoter → to express eukaryotic genes in the bacteria
 8. thus, a strong lac promoter is inserted next to an eukaryotic gene so that the bacteria transcription machineries , RNA polymerase can recognise and bind;

(b) Explain the steps in gel electrophoresis [7]

Agarose gel electrophoresis Is used for **resolving or separating DNA molecules** on the basis of **different molecular lengths or masses**

1. The gel is inserted into the electrophoresis chamber, and covered with an **electrophoretic buffer** which provide ions to support conductivity.
2. A **loading buffer with a tracking / marker dye** added to the DNA samples before loading into the sample wells. This serves to allow tracking of DNA that is colourless and allows visual monitoring of how far the electrophoresis has proceed.
3. The loading buffer contains glycerol, a dense liquid which helps the DNA sample to sink to bottom of well as denser.
4. Because of its phosphate groups, DNA is **negatively charged** at physiological / neutral pH. When electric current applied across the gel, opposite charges attract, DNA molecules must travel to reach the **positive electrode (anode)**.
5. The agarose comprises of a complex network of pores act like a sieve that separates molecules by molecular size → hence the migration distance is **inversely proportional** to the molecular size of a DNA fragment.
6. **DNA fragments (not visible in gel) are visualized by staining with ethidium bromide.** This fluorescent dye **intercalates** (penetrates) between the bases and when exposed to ultraviolet light, DNA fragments will appear as **discrete bands** on the gel.
7. **DNA markers** are loaded. The DNA markers serve as “ladders” of fragments of **known sizes** which is used to determine the size of DNA molecule studied.

- (c) Discuss the goals and benefits of the Human Genome Project. [7]

Goals

1. To identify all the approximate 20,000-25,000 genes in human DNA;
2. To determine the complete nucleotide sequence of the 3 billion chemical base pairs that make up entire human DNA;
3. To store this information in databases accessible to public;
4. To improve tools for data analysis;
5. To transfer related technologies for the private sector;
6. To address the ethical, legal and social issues that may arise from the project;

Benefits

Molecular medicine

- 1 Construct a detailed physical map of the entire genome to allow elucidation of genes associated with human diseases OR understanding the genetic basis of diseases;
- 2 Allows early diagnosis OR detection of genetic diseases; E.g. risk of breast cancer, diabetes and colon cancer;
- 3 Enhances research on pharmacogenomics: design customized drugs based on genetic profile. Drug design OR pharmacogenomics: 'custom drugs' → knowing about which genes affect a person's response to a drug OR genetic differences affect the way we react to a drug → possibility of tailoring drugs prescribed to fit a patient's pharmacogenomics profile for greater efficacy / to avoid dangerous side effects OR design of the most effective drug therapy and treatment strategies based on the specific genetic profile of a patient;
- 4 Pharmaceutical products such as medicine can be synthesized using readily available gene sequences, e.g. humulin or tissue plasminogen activator (TPA) given to after heart attack to dissolve blood clots.
- 5 Enables novel treatment through gene therapy as gene sequences are now readily available.
- 6 Allows genetic testing for the presence of a faulty gene for a condition that may not develop for some years, including having presymptomatic genetic testing for Huntington disease and predictive testing for young-onset forms of Alzheimer disease;
- 7 Allows large scale studies on genetically homogenous populations to identify alleles linked to certain disease conditions;

A. DNA Forensics

- 8 Identify potential suspects whose DNA may match evidence left at crime scenes;
- 9 Exonerate persons wrongly accused of crimes; /Identify crime and catastrophe victims
- 10 Establish paternity and other family relationships;
- 11 Match organ donors with recipients in transplant programs;

B. Bioarchaeology, anthropology, evolution and human migration

- 12 Study human evolution (through germline mutations in lineages);
- 13 Study of migration of different population groups based on female genetic inheritance OR lineage and migration of males via Y chromosomes;

C. Risk assessment

- 14 Provides a means of risk assessment to evaluate health risks of individuals who may be exposed to radiation or carcinogens